10. Number Sequences

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Going up by the same amount:

What comes next in the sequence: 23, 26, 29, 32, ?

Answer: Each number goes up by 3 (23, 26, 29, 32 = +3 each time). 32 + 3 = 35. Answer: **35**.

Example 2 — Going down by the same amount:

What comes next in the sequence: 54, 47, 40, 33, ?

Answer: Each number goes down by 7 (54, 47, 40, 33 = −7 each time). 33 − 7 = 26. Answer: **26**.

Example 3 — Multiplying each time:

What comes next in the sequence: 5, 10, 20, 40, ?

Answer: Each number is doubled (×2 each time). 40 × 2 = 80. Answer: **80**.

Example 4 — Multiple-choice:

What comes next in the sequence: 18, 25, 32, 39, ? A) 44 B) 45 C) 46 D) 47

Answer: The gap is +7 throughout: 18→25→32→39, so 39+7 = 46. Answer: **C) 46**.

Example 5 — Select-all:

For the sequence 60, 51, 42, 33, which of these are true? A) The gap between each number is 9. B) The next number would be 24. C) The sequence is increasing. D) The gap between 51 and 42 is 9. E) The first number is 60.

Answer: The gap is −9 the whole way through (A), so the next number is 33−9=24 (B); the gap between 51 and 42 is also 9 (D); 60 is the first number (E). The sequence goes down, not up, so C is false. Answer: **A, B, D and E**.

Foundation

15 questions

1. What comes next in the sequence: 21, 22, 23, 24, ?

2. What comes next in the sequence: 33, 34, 35, 36, ?

3. What comes next in the sequence: 45, 46, 47, 48, ?

4. What comes next in the sequence: 52, 53, 54, 55, ?

5. What comes next in the sequence: 19, 18, 17, 16, ?

6. What comes next in the sequence: 27, 26, 25, 24, ?

7. What comes next in the sequence: 40, 39, 38, 37, ?

8. What comes next in the sequence: 64, 65, 66, 67, ? A) 69 B) 68 C) 67 D) 66

9. What comes next in the sequence: 58, 57, 56, 55, ?

10. What comes next in the sequence: 71, 72, 73, 74, ?

11. What comes next in the sequence: 16, 15, 14, 13, ?

12. What comes next in the sequence: 29, 28, 27, 26, ?

13. Select ALL that apply: for the sequence 50, 48, 46, 44, which of these are true? A) The gap between each number is 2. B) The next number would be 42. C) The sequence is increasing. D) The gap between 48 and 46 is 2. E) The first number is 50.

14. What comes next in the sequence: 34, 35, 36, 37, ?

15. What comes next in the sequence: 62, 61, 60, 59, ?

Standard

20 questions

1. What comes next in the sequence: 3, 9, 15, 21, ?

2. What comes next in the sequence: 4, 11, 18, 25, ?

3. What comes next in the sequence: 50, 43, 36, 29, ?

4. What comes next in the sequence: 65, 57, 49, 41, ?

5. What comes next in the sequence: 12, 19, 26, 33, ? A) 40 B) 39 C) 41 D) 38

6. What comes next in the sequence: 7, 16, 25, 34, ?

7. What comes next in the sequence: 90, 80, 70, 60, ?

8. A number pattern always goes up by the same amount. The first 4 numbers are 11, 20, 29, 38. What is the next number in the pattern?

9. A number pattern always goes down by the same amount. The first 4 numbers are 84, 75, 66, 57. What is the next number in the pattern?

10. Select ALL that apply: for the sequence 18, 25, 32, 39, which of these are true? A) The gap between each number is 7. B) The next number would be 46. C) The sequence is decreasing. D) The gap between 25 and 32 is 7. E) The first number is 18.

11. What comes next in the sequence: 100, 89, 78, 67, ?

12. A number pattern always goes up by the same amount. The first 5 numbers are 6, 14, 22, 30, 38. What is the next number in the pattern?

13. What comes next in the sequence: 80, 71, 62, 53, ? A) 44 B) 45 C) 46 D) 47 E) 48

14. A number pattern always goes down by the same amount. The first 4 numbers are 77, 69, 61, 53. What is the next number in the pattern?

15. What comes next in the sequence: 2, 14, 26, 38, ?

16. What comes next in the sequence: 95, 84, 73, 62, ? A) 49 B) 50 C) 51 D) 52 E) 53

17. Select ALL that apply: for the sequence 85, 76, 67, 58, which of these are true? A) The gap between each number is 9. B) The next number would be 49. C) The sequence is increasing. D) The gap between 76 and 67 is 9. E) The first number is 85.

18. What comes next in the sequence: 9, 22, 35, 48, ?

19. A number pattern always goes up by the same amount. The first 4 numbers are 15, 29, 43, 57. What is the next number in the pattern?

20. What comes next in the sequence: 110, 97, 84, 71, ? A) 56 B) 57 C) 58 D) 59 E) 60

Hard

5 questions

1. What comes next in the sequence: 3, 6, 12, 24, ?

2. What comes next in the sequence: 2, 6, 18, 54, ? A) 108 B) 162 C) 216 D) 81

3. What comes next in the sequence: 320, 160, 80, 40, ?

4. Select ALL that apply: for the sequence 5, 10, 20, 40, which of these are true? A) Each number is double the one before. B) The next number would be 80. C) The sequence goes down. D) 20 is double 10. E) The first number is 5.

5. What comes next in the sequence: 4, 12, 36, 108, ?

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